Mathlearn · 6-sinf · Matematika

Geometrik shakllardagi noma'lum burchaklarni topish

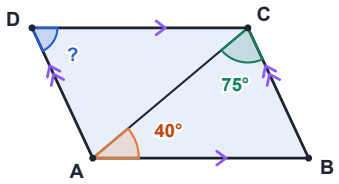
2-test (qiyinroq) · 20 ta savol

Ism-familiya: _____ Sinf: _____ Sana: _____

Har savolda bitta to'g'ri javobni belgilang.

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1. **ABCD — parallelogramm, AC — diagonal. Belgilangan burchak necha gradus?**

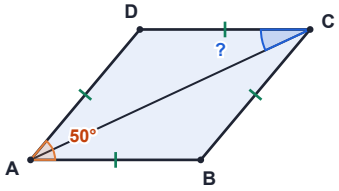


A) 75° **B)** 40°

C) 65° **D)** 115°

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2. **ABCD — romb, AC — diagonal. Belgilangan burchak necha gradus?**

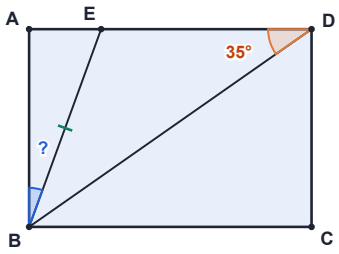


A) 65° **B)** 50°

C) 130° **D)** 25°

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3. **ABCD — to'g'ri to'rtburchak, BD — diagonal, E nuqta AD tomonda, BE = DE. Belgilangan burchak necha gradus?**

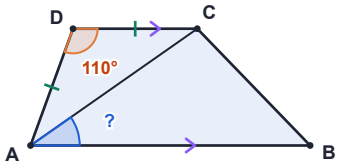


A) 55° **B)** 70°

C) 20° **D)** 35°

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4. **ABCD — trapetsiya, AB va DC — asoslar, AD = DC. Belgilangan burchak necha gradus?**



A) 45° **B)** 35°

C) 55° **D)** 70°

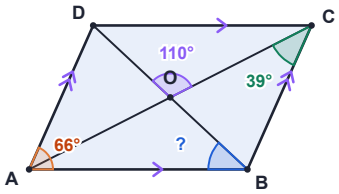
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5. **Ikki to'g'ri chiziq kesishdi. Bir to'g'ri chiziqda yonma-yon turgan ikki burchakdan biri ikkinchisidan 40° katta. Kattasi necha gradus?**

A) 220° **B)** 110° **C)** 70° **D)** 140°

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6. **ABCD — parallelogramm, uning diagonallari O nuqtada kesishadi. Belgilangan burchak necha gradus?**



A) 71° **B)** 43°

C) 66° **D)** 39°

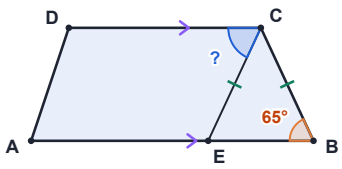
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7. **Teng yonli uchburchakning teng tomonlari orasidagi burchagi qolgan har bir burchagidan 30° katta. Shu qolgan burchak necha gradus?**

A) 50° **B)** 60° **C)** 80° **D)** 75°

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8. **ABCD — trapetsiya, AB va DC — asoslar. E nuqta AB asosda, CE = CB. Belgilangan burchak necha gradus?**



A) 130° **B)** 65°
C) 50° **D)** 115°

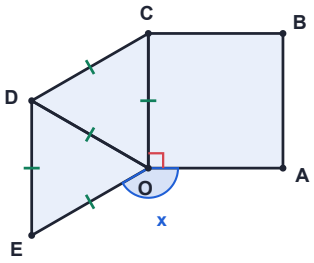
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9. **ABC uchburchakning BC tomoni D gacha davom ettirilgan, ∠ACD = 130° va ∠A = ∠B. ∠A necha gradus?**

A) 65° **B)** 25° **C)** 50° **D)** 130°

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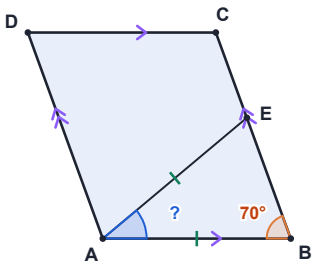
10. **O nuqtada kvadrat OABC va ikkita teng tomonli uchburchak OCD, ODE uchrashadi. x necha gradus?**



A) 210° **B)** 90°
C) 150° **D)** 120°

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11. **ABCD — parallelogramm, E nuqta BC tomonda, AB = AE. Belgilangan burchak necha gradus?**



A) 70° **B)** 55°
C) 110° **D)** 40°

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12. **Trapetsiyada AB va DC — asoslar. ∠C ning kattaligi ∠B dan 2 marta katta. ∠C necha gradus?**

A) 120° **B)** 108° **C)** 60° **D)** 144°

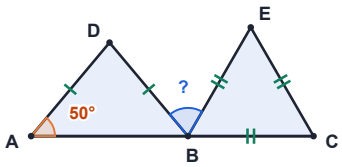
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13. **ABCD — trapetsiya, AB va DC — asoslar, ∠A = 60°, ∠B = 75°. Ali shunday yozdi: «∠C = ∠A = 60°, chunki qarama-qarshi burchaklar». ∠C aslida necha gradus?**

A) 120° **B)** 75° **C)** 105° **D)** 60°

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14. **A, B, C nuqtalar bitta to'g'ri chiziqda. ABD — teng yonli uchburchak (AD = BD), BCE — teng tomonli uchburchak. Belgilangan burchak necha gradus?**



A) 70° **B)** 50°
C) 80° **D)** 60°

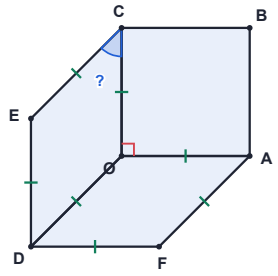
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15. **ABCD — kvadrat, E nuqta CD tomonda, ∠DAE = 18°. ∠AEC necha gradus?**

A) 18° **B)** 162° **C)** 72° **D)** 108°

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16. **O nuqta atrofida kvadrat OABC va ikkita bir xil romb OCED, ODFA bo'shliqsiz joylashgan. Belgilangan burchak necha gradus?**



A) 45° **B)** 90°
C) 60° **D)** 135°

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17. **ABC — teng tomonli uchburchak. BC tomon C dan davom ettirilib, D nuqta olingan, CD = CA. ∠BAD necha gradus?**

A) 120° **B)** 30° **C)** 60° **D)** 90°

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18. **ABCD — parallelogramm, BD — diagonal, ∠ABD = 90°, ∠A = 55°. ∠BDC necha gradus?**

A) 90° **B)** 35° **C)** 125° **D)** 55°

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19. **Uchburchakning bir tomoni davom ettirildi, hosil bo'lgan tashqi burchak 140°. Shu uchdan boshqa ikki burchakdan biri 75°. Ikkinchisi necha gradus?**

A) 65° **B)** 40° **C)** 75° **D)** 105°

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20. **ABCD — to'g'ri to'rtburchak, E nuqta AD tomonda. ∠ABE burchak ∠EBC dan 2 marta katta. ∠AEB necha gradus?**

A) 60° **B)** 90° **C)** 30° **D)** 120°

Javoblar

Geometrik shakllardagi noma'lum burchaklarni topish · 2-test (qiyinroq)

№	Javob	Qiymati
1	**C**	65°
2	**D**	25°
3	**C**	20°
4	**B**	35°
5	**B**	110°
6	**B**	43°
7	**A**	50°
8	**B**	65°
9	**A**	65°
10	**C**	150°

№	Javob	Qiymati
11	**D**	40°
12	**A**	120°
13	**C**	105°
14	**A**	70°
15	**D**	108°
16	**A**	45°
17	**D**	90°
18	**A**	90°
19	**A**	65°
20	**C**	30°

Qisqa yechimlar

1. ABC uchburchakda: ∠B = 180° − 40° − 75° = 65°.
∠D — ∠B ga qarama-qarshi: **65°**.

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2. ∠D = 180° − 50° = 130°.
AD = DC, demak ACD — teng yonli.
∠ACD = (180° − 130°) : 2 = **25°**.

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3. BED teng yonli (BE = DE): ∠EBD = ∠D = 35°.
BED uchburchakda ∠BED = 180° − 35° − 35° = 110°; AD to'g'ri chiziq: ∠AEB = 180° − 110° = 70°.
ABE uchburchakda ∠A = 90°: ∠ABE = 180° − 90° − 70° = **20°**.

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4. ACD teng yonli: ∠DAC = (180° − 110°) : 2 = 35°.
∠DAB = 180° − 110° = 70°.
∠CAB = 70° − 35° = **35°**.

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5. Ular bir to'g'ri chiziqda — yig'indisi 180°.
Kichigi: (180° − 40°) : 2 = 70°.
Kattasi: 70° + 40° = **110°**.

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6. ∠ABC = 180° − 66° = 114° (AB tomoniga yopishgan burchaklar).
∠AOB = ∠COD = 110° (vertikal); AC to'g'ri chiziq: ∠BOC = 180° − 110° = 70°.
BOC uchburchakda: ∠OBC = 180° − 70° − 39° = 71°.
∠ABD = 114° − 71° = **43°**.

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7. Qolgan burchaklar x va x, uchinchisi x + 30°.
3x + 30° = 180°, 3x = 150°, x = **50°**.

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8. CEB teng yonli (CE = CB): ∠CEB = ∠B = 65°, ∠ECB = 180° − 65° − 65° = 50°.
BC yon tomoni: ∠BCD = 180° − 65° = 115°.
∠DCE = 115° − 50° = **65°**.

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9. ∠ACB = 180° − 130° = 50°.
∠A + ∠B = 180° − 50° = 130°.
∠A = 130° : 2 = **65°**.

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10. O nuqtada: kvadrat 90°, ikki uchburchak 60° dan.
x = 360° − 90° − 60° − 60° = **150°**.

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11. ABE teng yonli (AB = AE): ∠AEB = ∠B = 70°.
∠BAE = 180° − 70° − 70° = **40°**.

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12. ∠B va ∠C — BC yon tomoniga yopishgan: ∠B + ∠C = 180°.
3 bo'lak = 180°, ∠B = 60°.
∠C = 2 · 60° = **120°**.

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13. Trapetsiya parallelogramm emas — qarama-qarshi burchaklari teng bo'lishi shart emas.
∠B va ∠C — BC yon tomoniga yopishgan: ∠C = 180° − 75° = **105°**.

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14. ABD teng yonli: ∠ABD = ∠A = 50°.
BCE teng tomonli: ∠CBE = 60°.
AC to'g'ri chiziq: ∠DBE = 180° − 50° − 60° = **70°**.

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15. ADE uchburchakda: ∠AED = 180° − 90° − 18° = 72°.
CD — to'g'ri chiziq: ∠AEC = 180° − 72° = **108°**.

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16. O nuqtada: 360° − 90° = 270° — ikki romb burchagi, har biri 270° : 2 = 135°.
Rombning OC tomoniga yopishgan burchaklar: ∠OCE = 180° − 135° = **45°**.

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17. ∠ACD = 180° − 60° = 120°.
ACD teng yonli: ∠CAD = (180° − 120°) : 2 = 30°.
∠BAD = 60° + 30° = **90°**.

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18. ABD uchburchakda: ∠ADB = 180° − 90° − 55° = 35°.
∠ADC = 180° − 55° = 125°.
∠BDC = 125° − 35° = **90°**.

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19. Shu uchdagi ichki burchak: 180° − 140° = 40°.
Qolgani: 180° − 40° − 75° = **65°**.

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20. ∠ABE + ∠EBC = 90° = 3 bo'lak, ∠EBC = 30°, ∠ABE = 60°.
ABE uchburchakda ∠A = 90°: ∠AEB = 180° − 90° − 60° = **30°**.
